

```
In [5]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [7]: df = pd.read_csv("Air bnb project.csv")
df
```

C:\Users\Ahsan\AppData\Local\Temp\ipykernel_6828\2916348628.py:1: DtypeWarning: Columns (25) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv("Air bnb project.csv")
```

Out[7]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	lon
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97236
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98236
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94997
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95412
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.95844
...	...	...	...	...	...	...	...	...	...
102594	6092437	Spare room in Williamsburg	12312296767	verified	Krik	Brooklyn	Williamsburg	40.70862	-73.95766
102595	6092990	Best Location near Columbia U	77864383453	unconfirmed	Mifan	Manhattan	Morningside Heights	40.80460	-73.95766

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	lon
102596	6093542	Comfy, bright room in Brooklyn	69050334417	unconfirmed	Megan	Brooklyn	Park Slope	40.67505	-73.75924
102597	6094094	Big Studio-One Stop from Midtown	11160591270	unconfirmed	Christopher	Queens	Long Island City	40.74989	-73.94925
102598	6094647	585 sf Luxury Studio	68170633372	unconfirmed	Rebecca	Manhattan	Upper West Side	40.76807	-73.98229

102599 rows × 26 columns

In [9]: df

Out[9]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	lon
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97235
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98225
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.95886
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95416
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73.95834
...	...	...	...	...	...	...	...	...	...
102594	6092437	Spare room in Williamsburg	12312296767	verified	Krik	Brooklyn	Williamsburg	40.70862	-73.95274
102595	6092990	Best Location near Columbia U	77864383453	unconfirmed	Mifan	Manhattan	Morningside Heights	40.80460	-73.95712

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long
102596	6093542	Comfy, bright room in Brooklyn	69050334417	unconfirmed	Megan	Brooklyn	Park Slope	40.67505	-73.9742
102597	6094094	Big Studio-One Stop from Midtown	11160591270	unconfirmed	Christopher	Queens	Long Island City	40.74989	-73.9500
102598	6094647	585 sf Luxury Studio	68170633372	unconfirmed	Rebecca	Manhattan	Upper West Side	40.76807	-73.9820

102599 rows × 26 columns

Check the column names in the Dataset

```
In [12]: df.columns
```

```
Out[12]: Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
               'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
               'country code', 'instant_bookable', 'cancellation_policy', 'room type',
               'Construction year', 'price', 'service fee', 'minimum nights',
               'number of reviews', 'last review', 'reviews per month',
               'review rate number', 'calculated host listings count',
               'availability 365', 'house_rules', 'license'],
              dtype='object')
```

Check for Missing Values

```
In [15]: df.isnull().sum()
```

```
Out[15]: id                                0
        NAME                             250
        host id                           0
        host_identity_verified             289
        host name                         406
        neighbourhood group               29
        neighbourhood                     16
        lat                               8
        long                              8
        country                           532
        country code                      131
        instant_bookable                  105
        cancellation_policy                76
        room type                          0
        Construction year                  214
        price                             247
        service fee                        273
        minimum nights                     409
        number of reviews                  183
        last review                        15893
        reviews per month                  15879
        review rate number                 326
        calculated host listings count     319
        availability 365                   448
        house_rules                        52131
        license                           102597
        dtype: int64
```

Handle Missing Values

""This code ensures that the 'last review' column is properly formatted as datetime, missing values in key columns are appropriately handled, and incomplete records are removed, preparing the dataset for further analysis or visualization ""

```
In [23]: # Convert 'last review' to datetime and handle errors
df['last review'] = pd.to_datetime(df['last review'], errors='coerce')
```

```
In [31]: # Fill missing values
df.fillna({'reviews per month': 0, 'last review': df['last review'].min()}, inplace=True)
```

```
In [33]: # Drop records with missing 'name' or 'host name'  
df.dropna(subset=['NAME', 'host name'], inplace=True)
```

```
In [35]: print(df.isnull().sum())
```

id	0
NAME	0
host id	0
host_identity_verified	276
host name	0
neighbourhood group	26
neighbourhood	16
lat	8
long	8
country	526
country code	122
instant_bookable	96
cancellation_policy	70
room type	0
Construction year	200
price	239
service fee	268
minimum nights	403
number of reviews	182
last review	0
reviews per month	0
review rate number	314
calculated host listings count	318
availability 365	420
house_rules	51867
license	101947

dtype: int64

Correct Data Types

```
In [38]: # Remove dollar signs and convert to float  
df['price'] = df['price'].replace('[\$',]', '', regex=True).astype(float)  
df['service fee'] = df['service fee'].replace('[\$',]', '', regex=True).astype(float)
```

```
<>:2: SyntaxWarning: invalid escape sequence '\$'  
<>:3: SyntaxWarning: invalid escape sequence '\$'  
<>:2: SyntaxWarning: invalid escape sequence '\$'  
<>:3: SyntaxWarning: invalid escape sequence '\$'  
C:\Users\Ahsan\AppData\Local\Temp\ipykernel_6828\138066982.py:2: SyntaxWarning: invalid escape sequence '\$'  
    df['price'] = df['price'].replace('[\$,]', '', regex=True).astype(float)  
C:\Users\Ahsan\AppData\Local\Temp\ipykernel_6828\138066982.py:3: SyntaxWarning: invalid escape sequence '\$'  
    df['service fee'] = df['service fee'].replace('[\$,]', '', regex=True).astype(float)
```

Remove Duplicates

```
In [47]: df.drop_duplicates(inplace=True)
```

Confirm Data Cleaning

```
In [50]: df.info()
```



```

<class 'pandas.core.frame.DataFrame'>
Index: 101410 entries, 0 to 102057
Data columns (total 26 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   id                                    101410 non-null  int64
 1   NAME                                101410 non-null  object
 2   host id                              101410 non-null  int64
 3   host_identity_verified               101134 non-null  object
 4   host name                            101410 non-null  object
 5   neighbourhood group                 101384 non-null  object
 6   neighbourhood                       101394 non-null  object
 7   lat                                 101402 non-null  float64
 8   long                                101402 non-null  float64
 9   country                             100884 non-null  object
10   country code                        101288 non-null  object
11   instant_bookable                    101314 non-null  object
12   cancellation_policy                 101340 non-null  object
13   room type                           101410 non-null  object
14   Construction year                   101210 non-null  float64
15   price                               101171 non-null  float64
16   service fee                         101142 non-null  float64
17   minimum nights                      101016 non-null  float64
18   number of reviews                   101228 non-null  float64
19   last review                         101410 non-null  datetime64[ns]
20   reviews per month                  101410 non-null  float64
21   review rate number                  101103 non-null  float64
22   calculated host listings count      101092 non-null  float64
23   availability 365                    100990 non-null  float64
24   house_rules                         49831 non-null  object
25   license                             2 non-null      object
dtypes: datetime64[ns](1), float64(11), int64(2), object(12)
memory usage: 20.9+ MB

```

```

In [52]: df = df.drop(columns=["license", "house_rules"], errors='ignore')
df

```

Out[52]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem	40.80902	-73
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem	40.79851	-73
5	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	Murray Hill	40.74767	-73
...	...	...	...	...	...	...	...	...	...
102053	57365208	Cozy bright room near Prospect Park	77326652202	unconfirmed	Mariam	Brooklyn	Flatbush	40.64945	-73
102054	57365760	Private Bedroom with Amazing Rooftop View	45936254757	verified	Trey	Brooklyn	Bushwick	40.69872	-73
102055	57366313	Pretty Brooklyn One-Bedroom for 2 to 4 people	23801060917	verified	Michael	Brooklyn	Bedford-Stuyvesant	40.67810	-73

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	
102056	57366865	Room & private bathroom in historic Harlem	15593031571	unconfirmed	Shireen	Manhattan	Harlem	40.81248	-73
102057	57367417	Rosalee Stewart	93578954226	verified	Stanley	Manhattan	Harlem	40.81315	-73

101410 rows × 24 columns

Descriptive Statistics

```
In [57]: df.describe()
```

Out[57]:

	id	host id	lat	long	Construction year	price	service fee	minimum nights
count	1.014100e+05	1.014100e+05	101402.000000	101402.000000	101210.000000	101171.000000	101142.000000	101016.000000
mean	2.920959e+07	4.926155e+10	40.728082	-73.949663	2012.486908	625.381008	125.043998	8.113744
min	1.001254e+06	1.236005e+08	40.499790	-74.249840	2003.000000	50.000000	10.000000	-1223.000000
25%	1.507574e+07	2.459183e+10	40.688730	-73.982570	2007.000000	340.000000	68.000000	2.000000
50%	2.922911e+07	4.912069e+10	40.722300	-73.954440	2012.000000	625.000000	125.000000	3.000000
75%	4.328308e+07	7.399747e+10	40.762750	-73.932340	2017.000000	913.000000	183.000000	5.000000
max	5.736742e+07	9.876313e+10	40.916970	-73.705220	2022.000000	1200.000000	240.000000	5645.000000
std	1.626820e+07	2.853703e+10	0.055850	0.049474	5.765130	331.609111	66.313374	30.378014



Visualization Distribution of Prices

In [60]: `# Plot the Distribution of Listing price`

```
In [70]: plt.figure(figsize=(10, 6))
sns.histplot(df['price'], bins=50, kde=True, color='red') # Set histogram color
plt.title('Distribution of Listing Prices')
plt.xlabel('Price ($)')
plt.ylabel('Frequency')
plt.show()
```

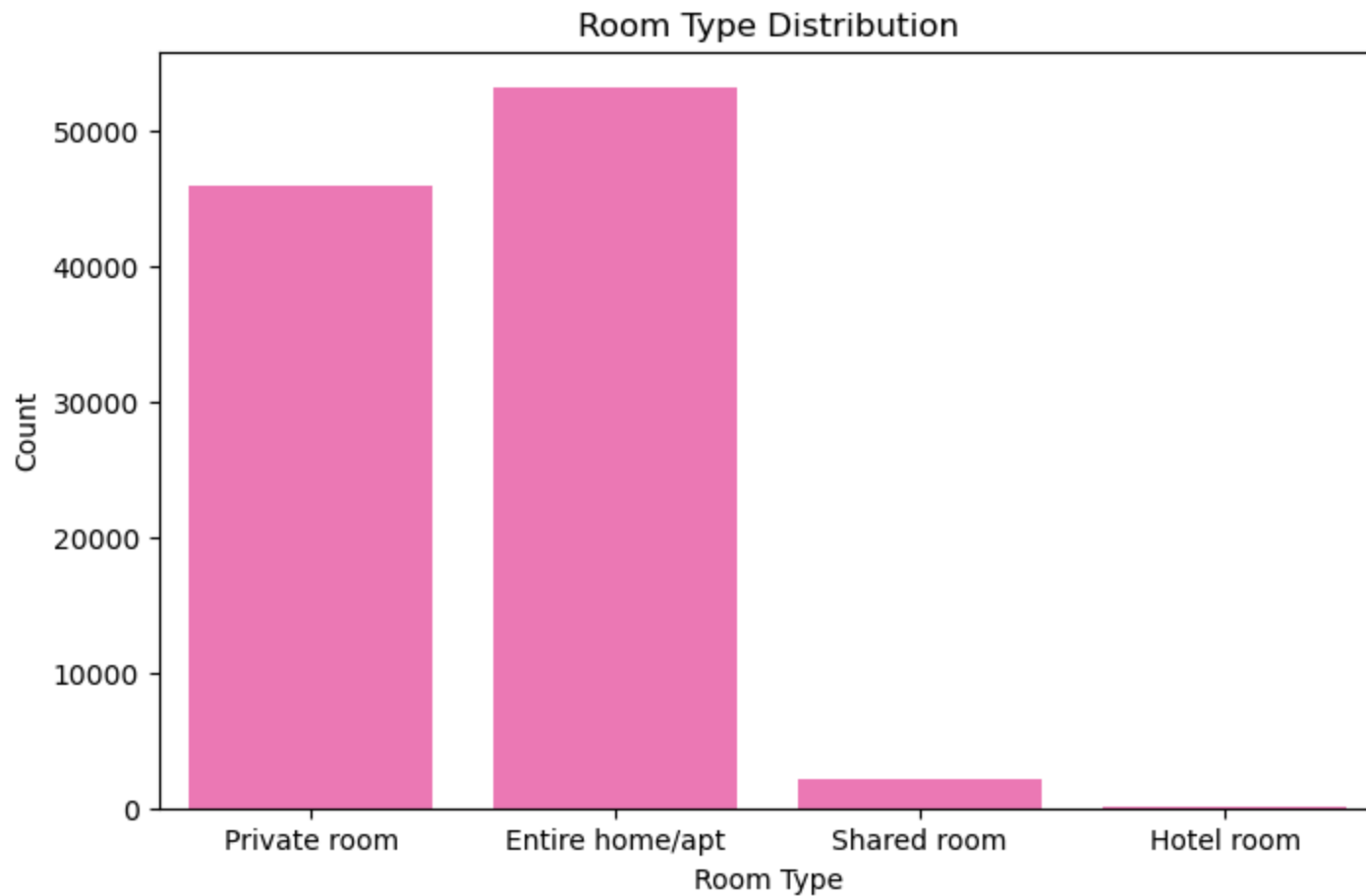


```
In [74]: # The histogram shows a fairly even distribution of  
# listing prices across different price ranges, indicating no  
# particular concentration of listings in any specific price range.  
# The KDE line helps visualize this even spread more clearly,  
# confirming that the dataset contains listings with a wide variety of prices
```

Room Type Analysis

```
In [77]: # Analyze the distribution of different room types
```

```
In [79]: plt.figure(figsize=(8, 5))  
sns.countplot(x='room type', data=df , color='hotpink')  
plt.title('Room Type Distribution')  
plt.xlabel('Room Type')  
plt.ylabel('Count')  
plt.show()
```

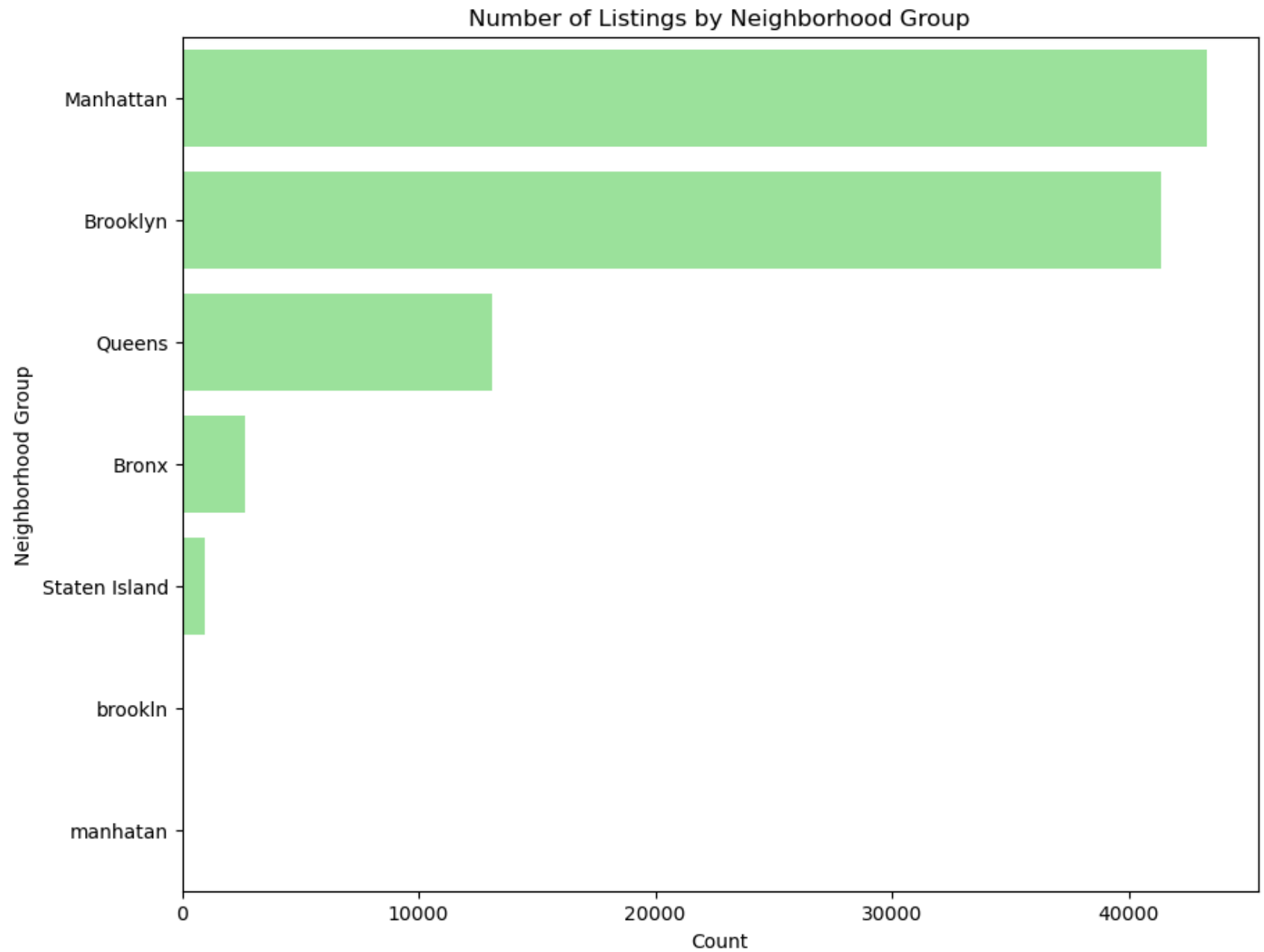


```
In [81]: '''The count plot shows a clear distribution of the different
room types available in the Airbnb dataset.
The majority of listings are for
'Entire home/apt' and 'Private room', with 'Shared room' and 'Hotel room'
being much less common. This insight can be useful
for understanding the availability and popularity of
different types of accommodations on Airbnb'''
```

```
Out[81]: "The count plot shows a clear distribution of the different\nroom types available in the Airbnb dataset.\nThe majori
ty of listings are for\n'Entire home/apt' and 'Private room', with 'Shared room' and 'Hotel room'\nbeing much less c
ommon. This insight can be useful\nfor understanding the availability and popularity of\ndifferent types of accommod
ations on Airbnb"
```

Neighborhood Analysis

```
In [105... plt.figure(figsize=(10, 8))
sns.countplot(y='neighbourhood group', data=df,color="lightgreen", order=df['neighbourhood group'].value_counts().:
plt.title('Number of Listings by Neighborhood Group')
plt.xlabel('Count')
plt.ylabel('Neighborhood Group')
plt.show()
```



```
In [ ]: '''The count plot shows a clear distribution  
of the number of listings across different neighborhood
```

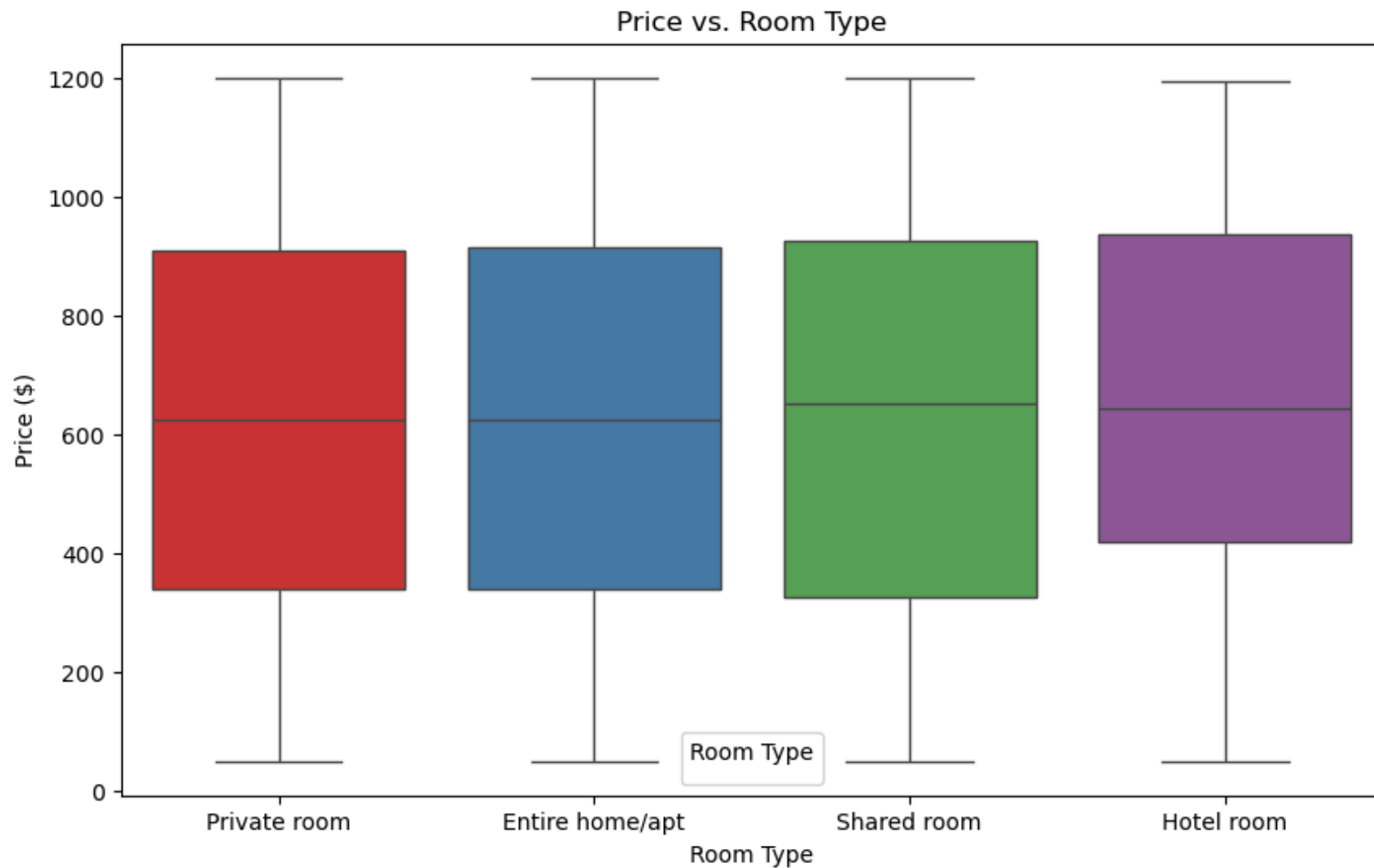

groups. Manhattan and Brooklyn dominate the listings, suggesting they are prime locations for Airbnb. Queens, Bronx, and Staten Island have fewer listings, indicating less availability or popularity.'''

Price vs. Room Type

In [108...

```
plt.figure(figsize=(10, 6))
sns.boxplot(x='room type', y='price', hue='room type', data=df, palette='Set1')
plt.title('Price vs. Room Type')
plt.xlabel('Room Type')
plt.ylabel('Price ($)')
plt.legend(title='Room Type')
plt.show()
```

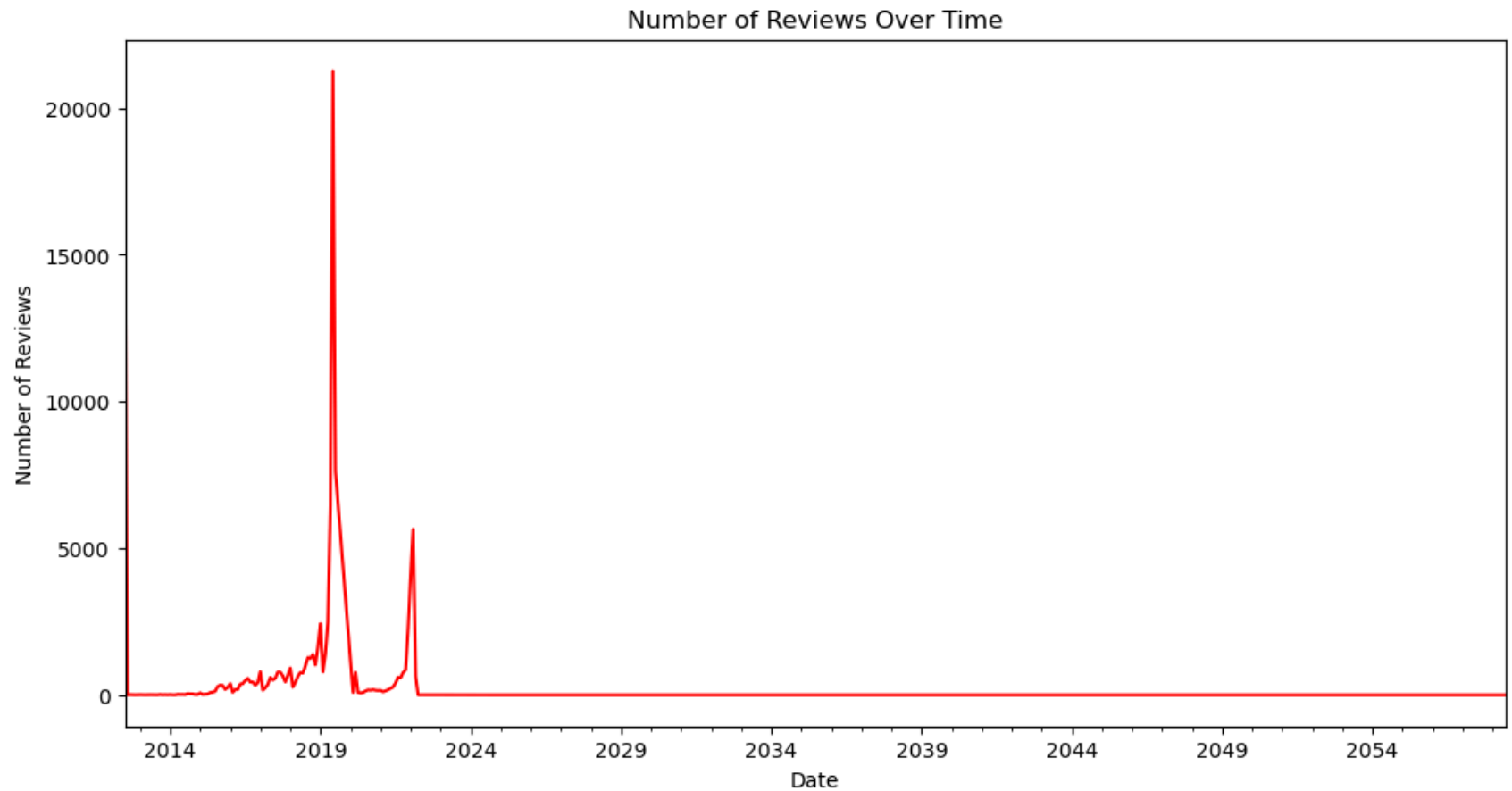
C:\Users\Ahsan\AppData\Local\Temp\ipykernel_6828\1262699844.py:6: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.
plt.legend(title='Room Type')



```
In [ ]: '''Price vs. Room Type
The box plot provides a detailed view
of how prices vary across different room types
in the Airbnb dataset. It shows that while
'Shared room' tends to have lower prices, 'Private room',
'Entire home/apt', and 'Hotel room' have higher and more varied price ranges.
This visualization helps in understanding the pricing
dynamics for different types of accommodations on Airbnb.'''
```

Reviews Over Time

```
In [111... df['last review'] = pd.to_datetime(df['last review'])
reviews_over_time = df.groupby(df['last review'].dt.to_period('M')).size()
plt.figure(figsize=(12, 6))
reviews_over_time.plot(kind='line', color='red')
plt.title('Number of Reviews Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Reviews')
plt.show()
```



```
In [ ]: The line plot provides a clear visualization
of the number of reviews over time.
It helps identify trends and patterns in review activity,
such as periods of high or low activity.
This information can be useful for understanding
the dynamics of user engagement and the popularity
of Airbnb listings over time. The significant spikes
and drops in reviews might be worth further investigation
to understand the underlying causes, such as changes
in Airbnb policies, market conditions, or external events.
```

```
In [135... df = pd.DataFrame(df)
```

```
In [141... # Save the DataFrame to a CSV file
df.to_csv("Air Bnb Data Analysis project.csv", index=False)
```

```
In [142... print("Air Bnb Data Analysis project.csv")
```

Air Bnb Data Analysis project.csv

```
In [ ]:
```